

1. Administrative & Study Identification

Full Study Title: A Phase Ib/III, Open-label, Randomised Study of Capivasertib plus CDK4/6 Inhibitors and Fulvestrant versus CDK4/6 Inhibitors and Fulvestrant in Hormone Receptor-Positive and Human Epidermal Growth Factor Receptor 2-Negative Locally Advanced, Unresectable or Metastatic Breast Cancer **Short Title/Acronym:** CAPitello-292 **Protocol Number:** D361DC00001 **NCT Number:** NCT04862663 **Trial Status:** Recruiting (Currently recruiting participants) **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Capivasertib (in combination with CDK4/6 inhibitors [palbociclib or ribociclib] and fulvestrant [Faslodex]) **Phase of Development:** Phase Ib/III (PHASE3) **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (Progression-Free Survival [PFS]) of adding capivasertib to fulvestrant and the investigator's choice of CDK4/6 inhibitors (either palbociclib or ribociclib) in patients with HR+/HER2- advanced breast cancer. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DoR), Clinical Benefit Rate (CBR), Progression-Free Survival in patients with AKT pathway-altered tumors (PIK3CA/AKT1/PTEN), and overall safety/tolerability.

2.2 Background & Rationale Overcoming resistance to endocrine therapy in advanced breast cancer (ABC) is a major challenge. AKT pathway activation is implicated in resistance to endocrine therapy and cyclin-dependent kinase 4/6 (CDK4/6) inhibitors in patients with HR+/HER2- ABC. This study aims to evaluate whether simultaneous inhibition of the AKT and CDK4/6 pathways with capivasertib may delay resistance or re-sensitize tumors, leading to improved clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (CDK4/6i + Fulvestrant) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target **SN\$:** 895 participants **Number of Sites:** Approximately 23 countries globally **Estimated Study Start:** 10-May-2021 **Estimated Primary Completion:** 01-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult females (pre-/peri-/ and post-menopausal) and adult males. 2. Histologically confirmed HR+/HER2- breast cancer determined from the most recent tumor sample (primary or metastatic). 3. Eligible for fulvestrant therapy and at least one of the following CDK4/6 inhibitors: palbociclib or ribociclib, following recurrence or progression on or within 12 months of completing a (neo)adjuvant ET regimen.

4.2 Exclusion Criteria 1. History of another primary malignancy (except for malignancy treated with curative intent with no known active disease ≥ 2 years before the first dose). 2. Clinically significant abnormalities of glucose metabolism at screening (e.g., diabetes mellitus type I or type II requiring insulin treatment; HbA1c $\geq 8.0\%$). 3. Radiotherapy within 2 weeks or major surgery/significant traumatic injury within 4 weeks of the first dose. 4. Active spinal cord compression, brain metastases, or leptomeningeal metastases (unless definitively treated and clinically stable). 5. Clinically significant cardiac criteria including QTcF $\geq 450/470$ ms, important abnormalities in rhythm/conduction, MI/angina/CABG within 6 months, uncontrolled hypo/hypertension, NYHA grade ≥ 2 heart failure, or LVEF $< 50\%$. 6. Previous allogeneic bone marrow or solid organ transplant. 7. Any prior treatment with AKT, PI3K, or mTOR inhibitors. 8. Prior treatment with CDK4/6 inhibitors in the metastatic setting, >1 line of chemotherapy for metastatic disease, or any line of endocrine-based therapy for inoperable locally advanced or metastatic disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Capivasertib:** Orally for 4 days followed by 3 days off per week. * **CDK4/6i (Palbociclib or Ribociclib):** Orally for 21 days followed by 7 days off per 28-day cycle. * **Fulvestrant (Trade Name: Faslodex, ATC Code: L02BA03):** Injection given once every 28 days (with one extra loading dose on Day 15 of the first cycle). **Route of Administration:** Oral (Capivasertib, CDK4/6i) and Intramuscular/Injection (Fulvestrant). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptoms and comorbidities. **Prohibited:** Concurrent anti-cancer therapy, other investigational agents, and prior use of PI3K/AKT/mTOR inhibitors.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor sample testing for PIK3CA/AKT1/PTEN status, Baseline Imaging
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 28 Days	Treatment dispensing, Safety Labs, Efficacy Assessment (Tumor scans per protocol timeline), Vital Signs
End of Study	Upon PD / Toxicity	Final physical exam, Exit Interview, IP Accountability, Transition to Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from randomization until objective disease progression or death due to any cause. **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) according to RECIST v1.1.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) in patients with AKT pathway-altered tumors * Overall Survival (OS) * Objective Response Rate (ORR) and Duration of Response (DoR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine evaluation of Dose-Limiting Toxicities (DLTs), AEs, clinical chemistry, hematology, glucose metabolism parameters, and ECG parameters. **Serious Adverse Events (SAEs):** Reported continuously throughout the treatment period and safety follow-up. **Data Monitoring Committee (DMC):** External Safety Review Committee (SRC) oversight to review dose-finding, tolerability, and overarching study safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Appropriately powered to detect a statistically significant improvement in PFS for the triplet combination versus the control arm in the overall population and the AKT pathway-altered subgroup. **Handling of Missing Data:** Censoring rules applied for missing progression events as per RECIST v1.1 regulatory guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard rigorous onsite and remote monitoring of trial centers. **Audit Trail:** eCRF locking, source data verification, and secure data clarification processes.

11. Study Identifiers & Governance

Primary ID: D361DC00001 **Posting ID:** 756877908927442163 **Registry Number:** NCT04862663 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Capivasertib + CDK4/6i + Fulvestrant for Advanced/Metastatic HR+/HER2- Breast Cancer (CAPitello-292) **Official Regulatory Title:** A Phase Ib/III, Open-label, Randomised Study of Capivasertib plus CDK4/6 Inhibitors and Fulvestrant versus CDK4/6 Inhibitors and Fulvestrant in Hormone Receptor-Positive and Human Epidermal Growth Factor Receptor 2-Negative Locally Advanced, Unresectable or Metastatic Breast Cancer

12. Clinical Framework (Disease Specific)

Disease Area: Locally Advanced (Inoperable) or Metastatic Breast Cancer **General Condition:** Breast Cancer (Malignant neoplasm of breast) **Preferred UMLS Name:** Hormone Receptor Positive HER2 Negative Breast Cancer **Diagnosis Threshold:** Histologically confirmed HR+/HER2- Status with disease progression/recurrence **Medical Methodology:** Targeted Combination Therapy (AKT inhibitor + CDK4/6 inhibitor + Endocrine Therapy) **Optimization Target:** Delayed treatment resistance and optimized Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Standard oncological pathway for HR+/HER2-metastatic disease **Education Protocol (HCP):** Protocol specific training, investigational product administration, and rigorous safety reporting **Education Protocol (Patient):** Informed consent details, specific dosing schedule adherence (days on/off), and side-effect recognition **Quality Audit Mechanism:** Source data verification & independent central review (BICR) of imaging

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity (Approx. 8 years estimated study span) **Onsite Visit Cadence:** Every 28 days (per treatment cycle) **Remote Monitoring Frequency:** Continuous evaluation corresponding with cycle-based

toxicity checks **Checklist Review Interval:** Tumor assessments continuously scheduled every 8 to 12 weeks per standard of care

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					